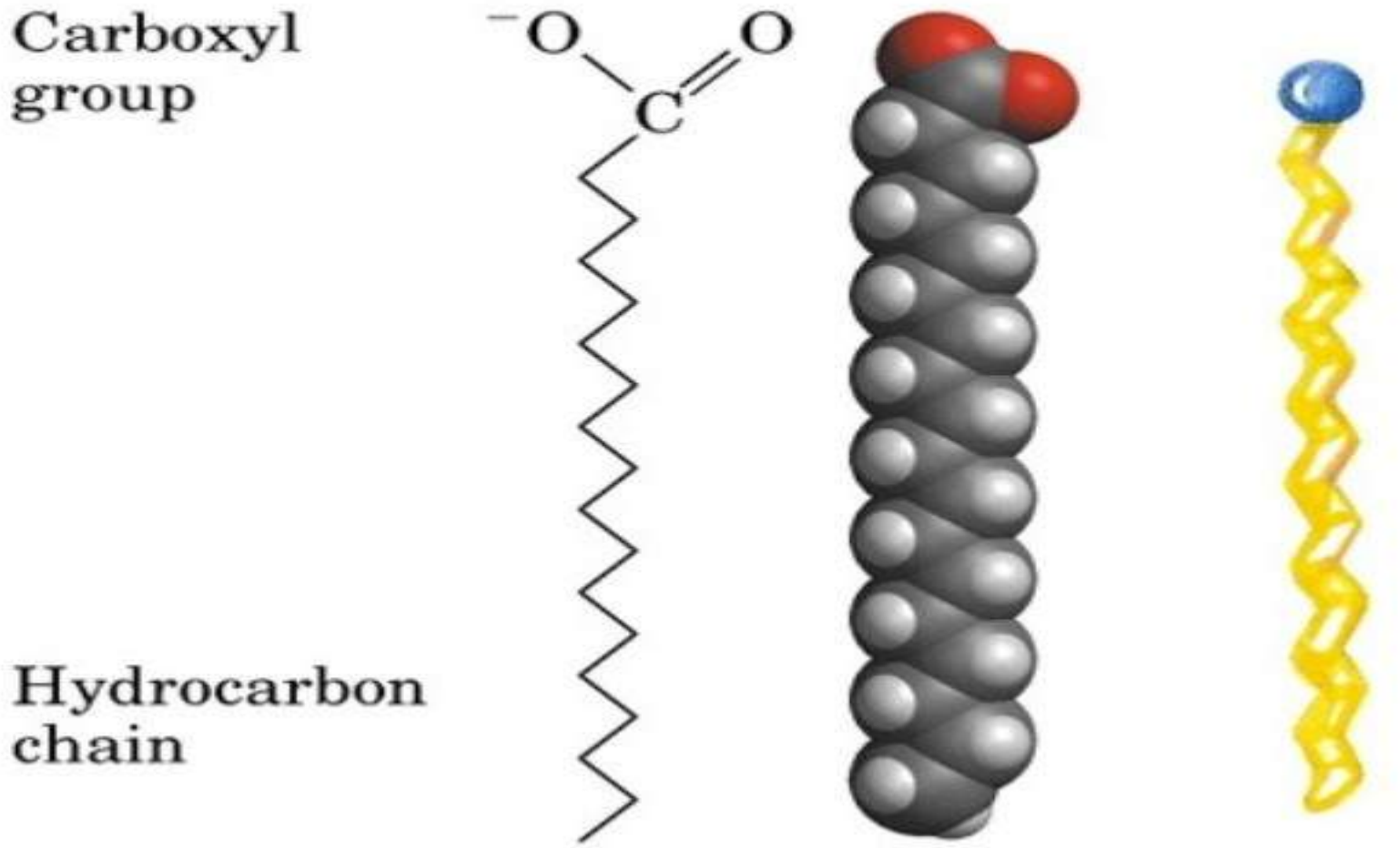
Fatty acids: Structure, Classification and Functions



Learning Objectives

At the end of this session students will be able to understand;

1. Lipids, Fatty acids, and its classification.
2. Structure and roles of fatty acids.
3. Biological roles of various fatty acids.

- The lipids constitute a very important heterogeneous group of organic substances in plant and animal tissues, and ***related either actually or potentially to the fatty acids.***

- Chemically they are various types of esters of different alcohols. **In addition to alcohol and fatty acids, some of the lipids may contain phosphoric acid, nitrogenous base and carbohydrates.**

Bloor’s Criteria

According to **Bloor**, lipids are compounds having the following characteristics:

- They are ***insoluble in water.***

- Solubility in one or more organic solvents, such as ether, chloroform, benzene, acetone, etc, so called ***fat solvents.***

- Some relationship to the fatty acids as esters either actual or potential.

- Possibility of utilization by living organisms.

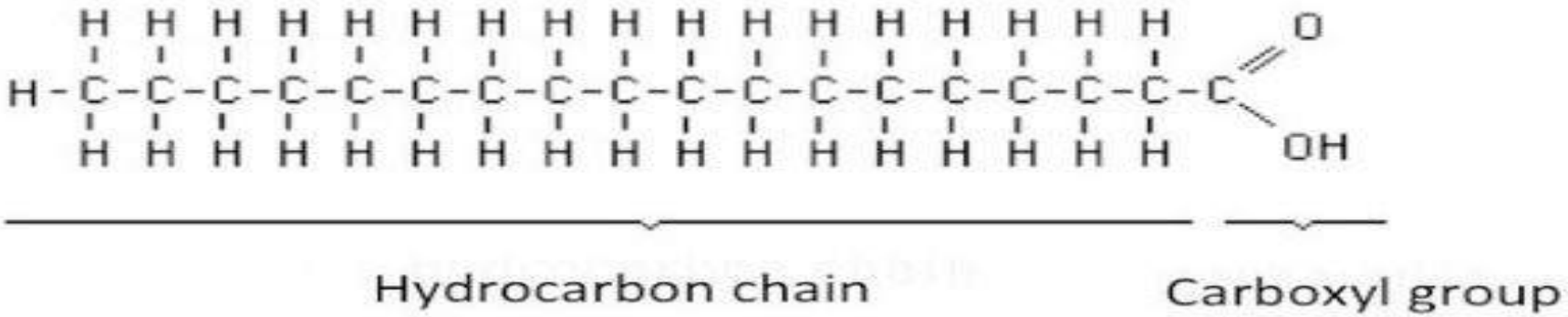
Introduction

- ***Functions***

  - Energy storage
  - Structure of cell membranes
  - Thermal blanket and cushion
  - Precursors of hormones (steroids and prostaglandins)

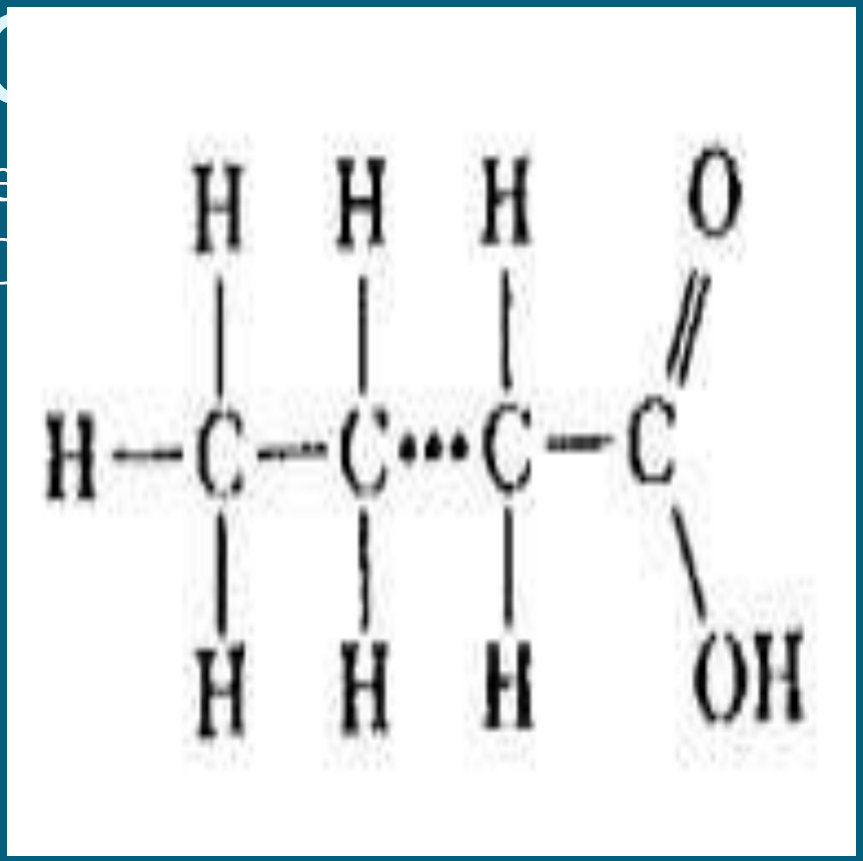
Fatty acids

- Fatty acids are a class of compounds containing a long hydrophobic hydrocarbon chain and a terminal carboxylate group
- They exist free in the body as well as fatty acyl esters in more complex molecules such as triglycerides or phospholipids.
- Fatty acids can be oxidized in all tissues, particularly liver and muscle to provide energy
- They are also structural components of membrane lipids such as phospholipids and glycolipids.
- Esterified fatty acids, in the form of triglycerides are stored in adipose cells



STRUCTURE OF FATTY AC

- A fatty acid is nothing more chain with a carboxyl group (C



- The COOH gives it an acid property.
- The 3….dots represent the chain is very long.

Fatty Acid (%)	Composition	Chain Fatty Acid	Common Source
Butyric Acid	C4:0	Short Chain Fatty Acid	Butterfat
Caproic Acid	C6:0	Short Chain Fatty Acid	Butterfat
Caprylic Acid	C8:0	Medium Chain Fatty Acid	Coconut Oil/Palm Oil
Capric Acid	C10:0	Medium Chain Fatty Acid	Coconut Oil/Palm Oil
Lauric Acid	C12:0	Medium Chain Fatty Acid	Coconut Oil/Palm Oil
Myristic Acid	C14:0	Long Chain Fatty Acid	Nutmeg Oil
Palmitic Acid	C16:0	Long Chain Fatty Acid	Animal & Vegetable Oil
Stearic Acid	C18:0	Long Chain Fatty Acid	Animal & Vegetable Oil
Oleic Acid	C18:1	Long Chain Fatty Acid	Olive Oil
Linoleic Acid	C18:2	Long Chain Fatty Acid	Vegetable Oil
Linolenic Acid	C18:3	Long Chain Fatty Acid	Linseed Oil

- There are more than 100 different FAs in nature and most FAs have an *even number of carbon atoms* (mainly due to the mechanism of biosynthesis-the concatenation of C2 units)

- Mainly present in the form of *esters (not free)*

- In higher plants and animals, the predominant FA residues are those of the *C16 and C18 species*(palmitic, oleic, linoleic, and stearic acids).

- FAs with less than 14 and more than 20 carbonatoms are uncommon

Properties

Most of fatty acids - (in biology) have an even number of C-atom.

Usually chain length – 14 to 24 C-atoms

Biological properties of fatty acids mainly depends on their chain length and the number of double bonds.

Shorter chain length fatty acids have lower melting temperature than those with longer chain.

Properties

Saturated fatty acids are solid at room temperature and have a high melting point

Unsaturated fatty acids are liquid at room temperature and have a low melting point

Fatty Acids

- The most common fatty acids.

	Carbon Atoms: Double Bonds	Common Name	Melting Point (°C)
	Saturated Fatty Acids		
Higher mp	12:0	Lauric acid	44
	14:0	Myristic acid	58
	16:0	Palmitic acid	63
	18:0	Stearic acid	70
	20:0	Arachidic acid	77
	Unsaturated Fatty Acids		
Lower mp	16:1	Palmitoleic acid	1
	18:1	Oleic acid	16
	18:2	Linoleic acid	-5
	18:3	Linolenic acid	-11
	20:4	Arachidonic acid	-49

Short chain saturated fatty acids

- Bad smell
- Fluids
- Present in milk fat (C4-C10)
- Easily digestible

Long chain saturated fatty acids

- Solid at room temperature
- Most common in animal and plant fats
- Less easily digestible

Classification

CLASSIFICATION OF FAs

- **According to the chain length**

• *short-chain fatty acid* - SCFA	< 6 carbon atoms
• *medium-chain fatty acid* - MCFA	6-12 carbon atoms
• *long-chain fatty acid* - LCFA	14-20 carbon atoms
• *very-long chain fatty acid* - VLCFA	> 20 carbon atoms

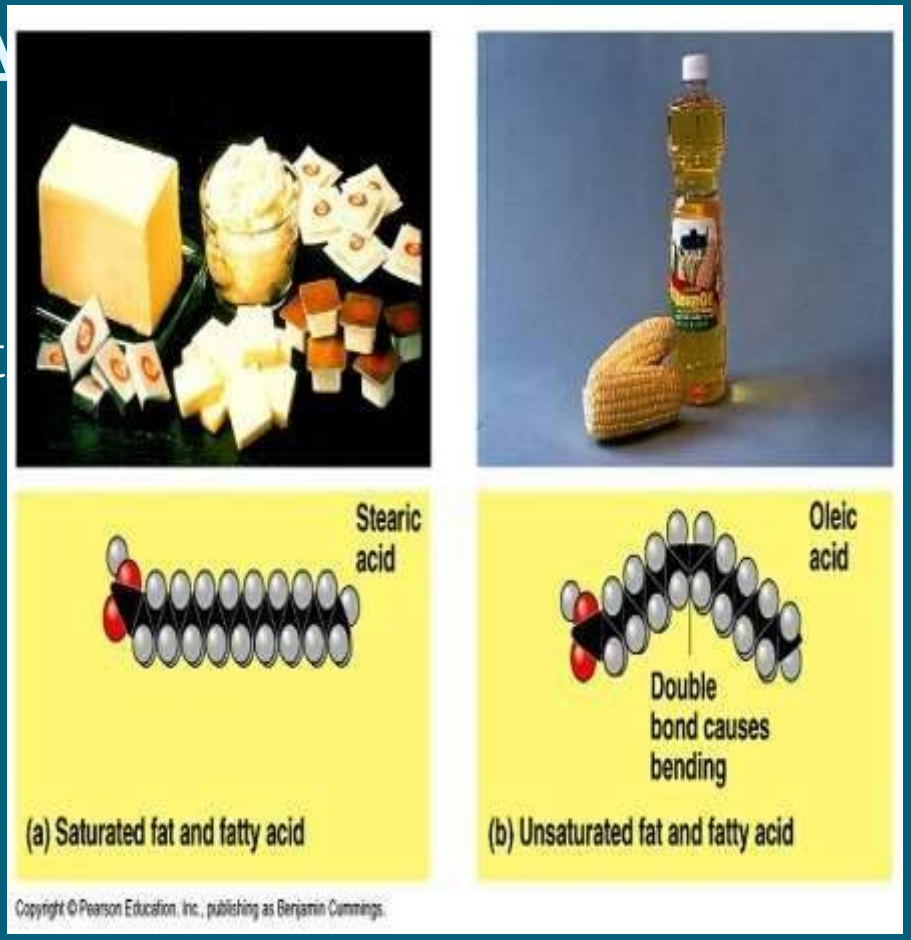
- **According to the degree of saturation** (presence or absence of double bonds)
  - *saturated*
  - *unsaturated cis/trans isomers*

- **Branched** (unusual)
- **Hydroxylated** (unusual)

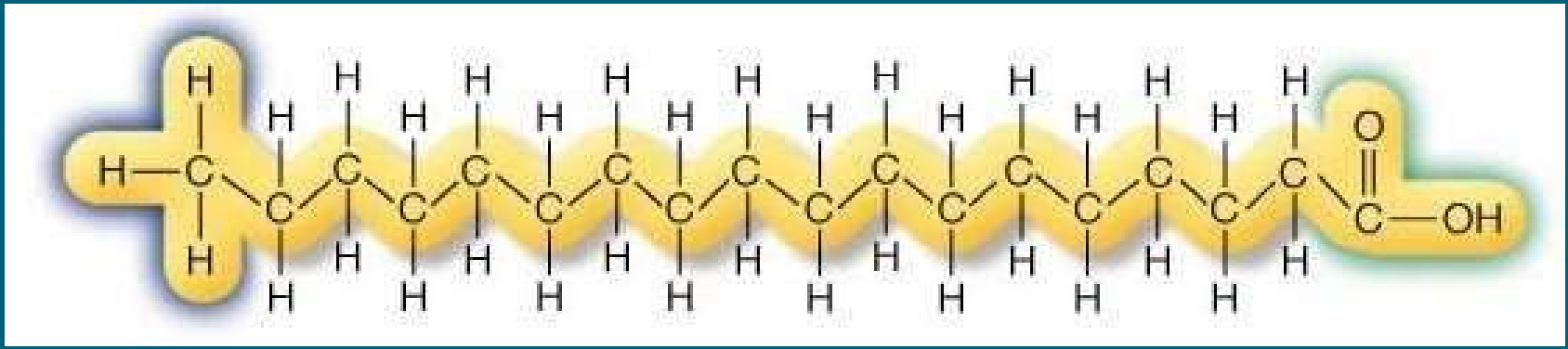
SATURATED & UNSATURATED F. A

- Saturated FAs have no double bonds lower the melt

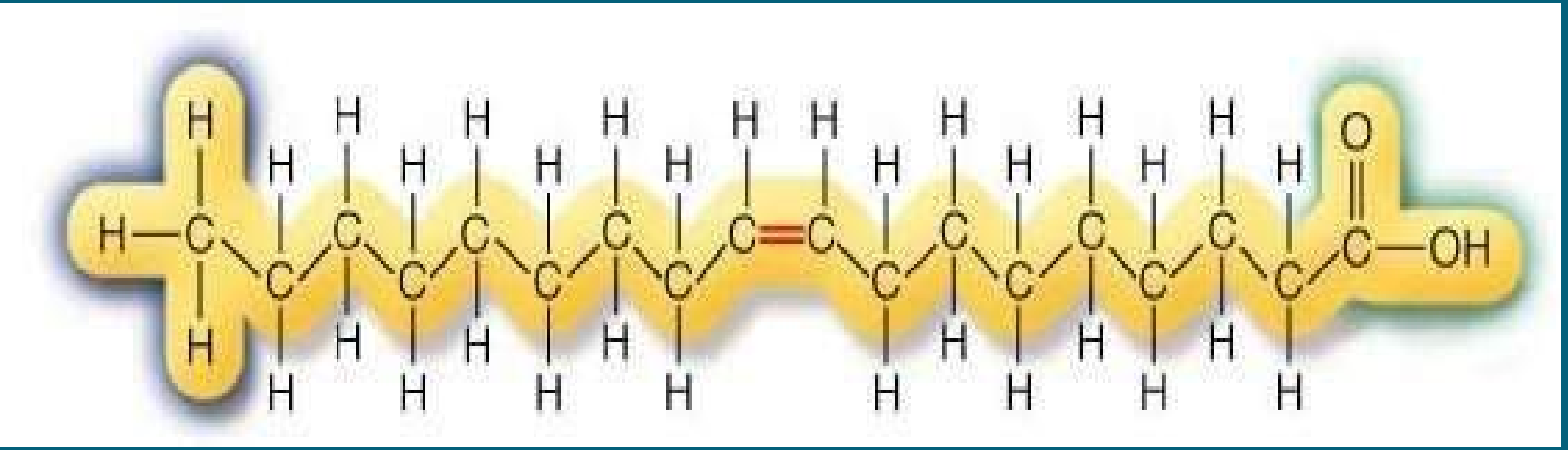
- Unsaturated FAs have at least one double bond (C=C) in one of the fatty acids



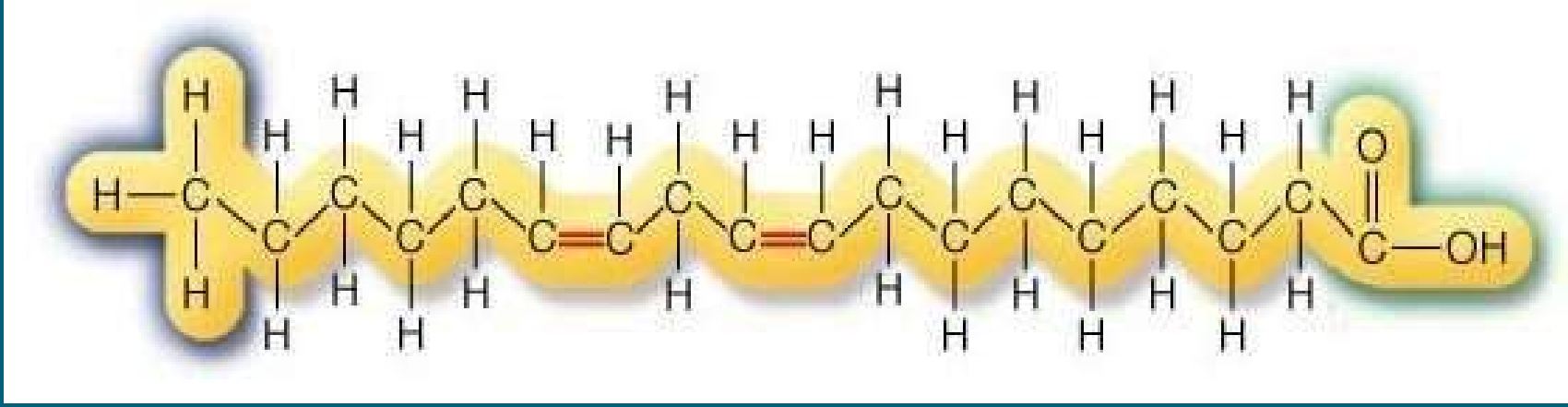
SATURATED FATTY ACIDS



MONOSATURATED FATTY ACIDS



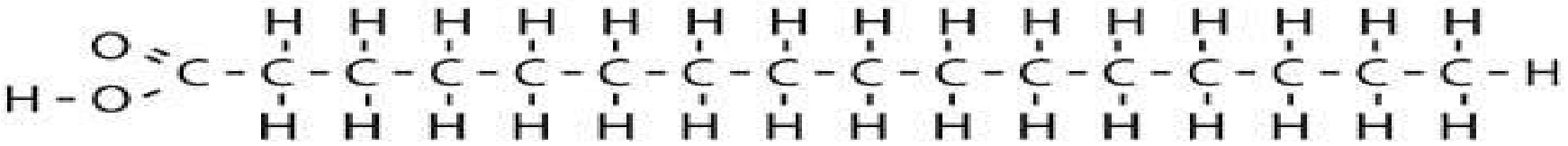
POLYSATURATED FATTY ACIDS



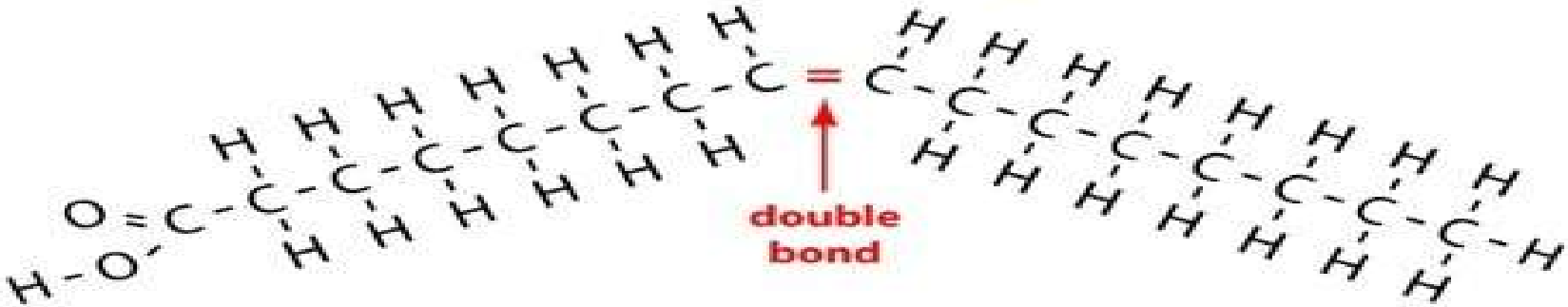
LOCATION OF DOUBLE BOND

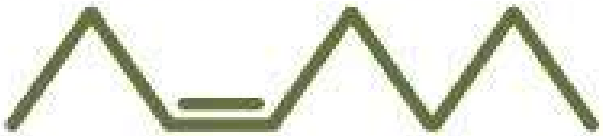
- Polyunsaturated fatty acid (PUFA) are identified by position of the double bond nearest the methyl end (CH_3) of the carbon chain; this is described as a omega number.
- If PUFA has first double bond 3 carbons away from the methyl end => omega 3 FA
- 6 carbons from methyl end => omega 6 FA

saturated fatty acid

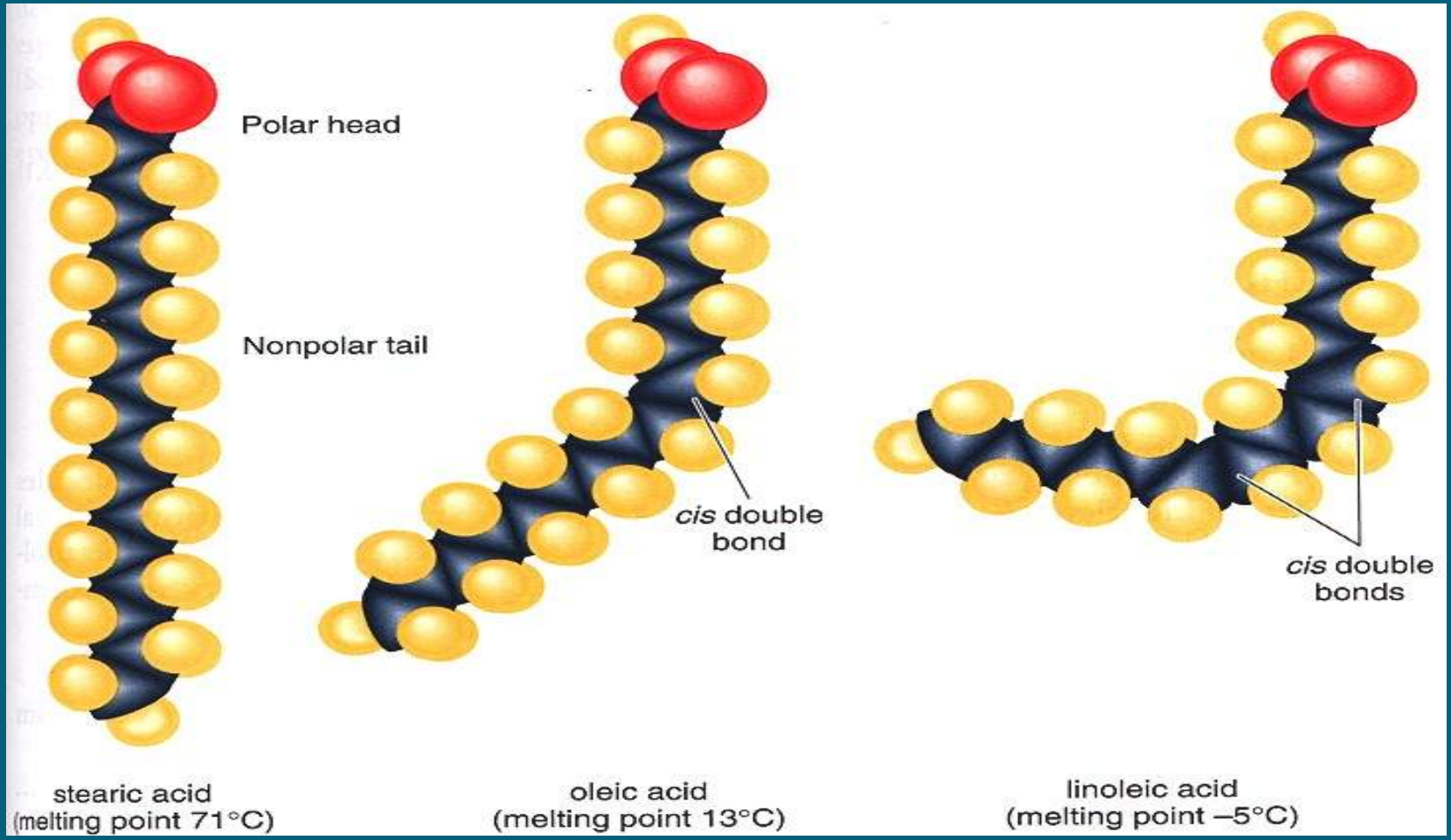


unsaturated fatty acid



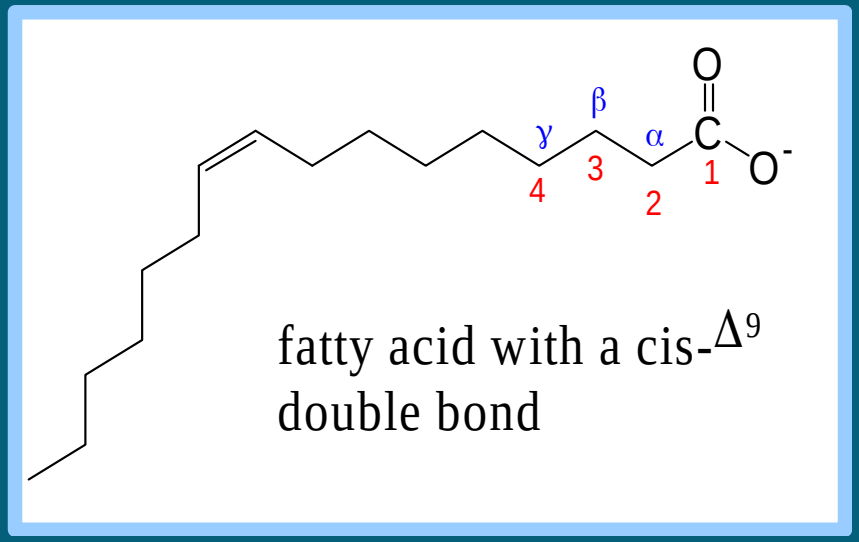
Type of Fatty Acid	Double Bonds	Diagram
Saturated	None	
Monounsaturated	One	
Polyunsaturated	Multiple (>1)	

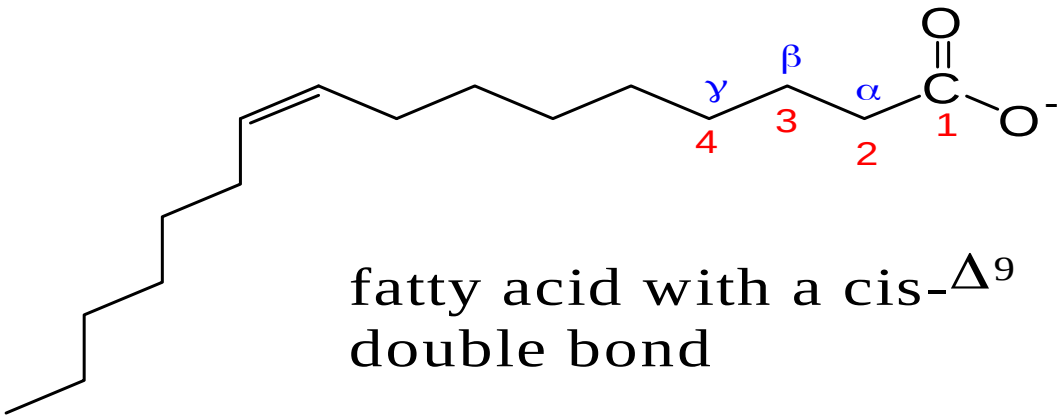
Fatty Acids


Polar head
Nonpolar tail
cis double bond
cis double bonds
stearic acid
(melting point 71°C)
oleic acid
(melting point 13°C)
linoleic acid
(melting point –5°C)

Double bonds in fatty acids usually have the cis configuration.

Most naturally occurring fatty acids have an **even number** of carbon atoms.





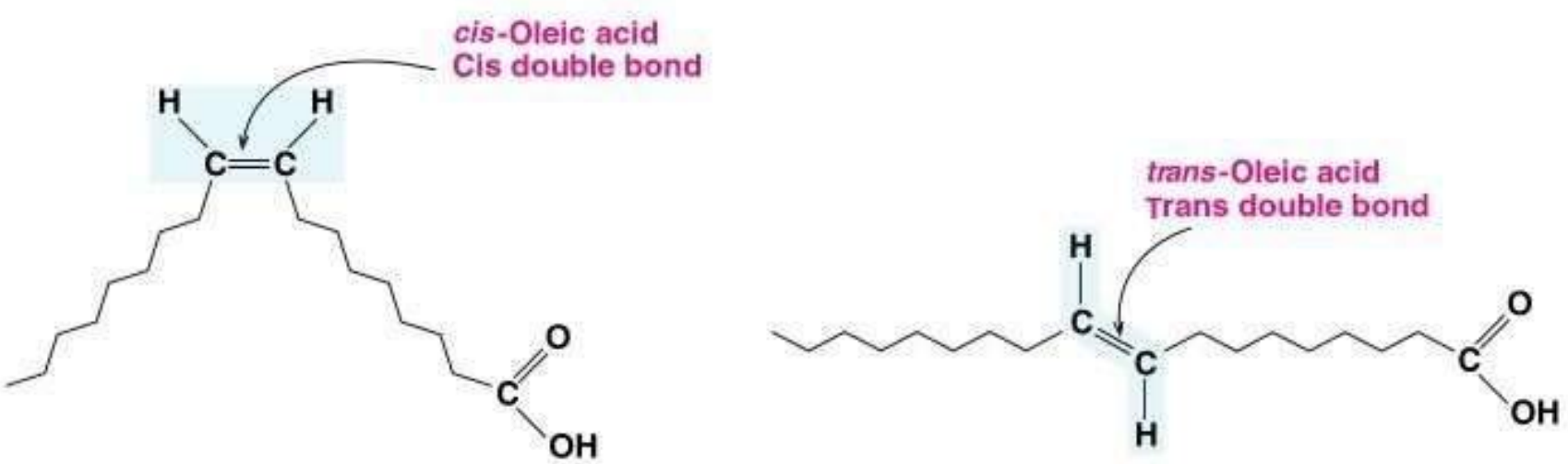
There is free rotation about **C-C** bonds in the fatty acid hydrocarbon, except where there is a double bond.

Each cis double bond causes a **kink** in the chain. Rotation about other **C-C** bonds would permit a more linear structure than shown, but there would be a kink.

Cis and Trans Unsaturated Fatty Acids

- Natural unsaturated fatty acids have cis double bonds
- When unsaturated vegetable oils are hydrogenated to form more saturated oils (as in margarine), some of the cis fatty acids are isomerized to trans fatty acids

- Trans fatty acids are much more linear than cis fatty acids, so their melting points are higher and studies have shown that trans fats may act similarly to saturated fats and could contribute to heart disease and some cancers



Timberlake, *General, Organic, and Biological Chemistry*. Copyright © Pearson Education Inc., publishing as Benjamin Cummings

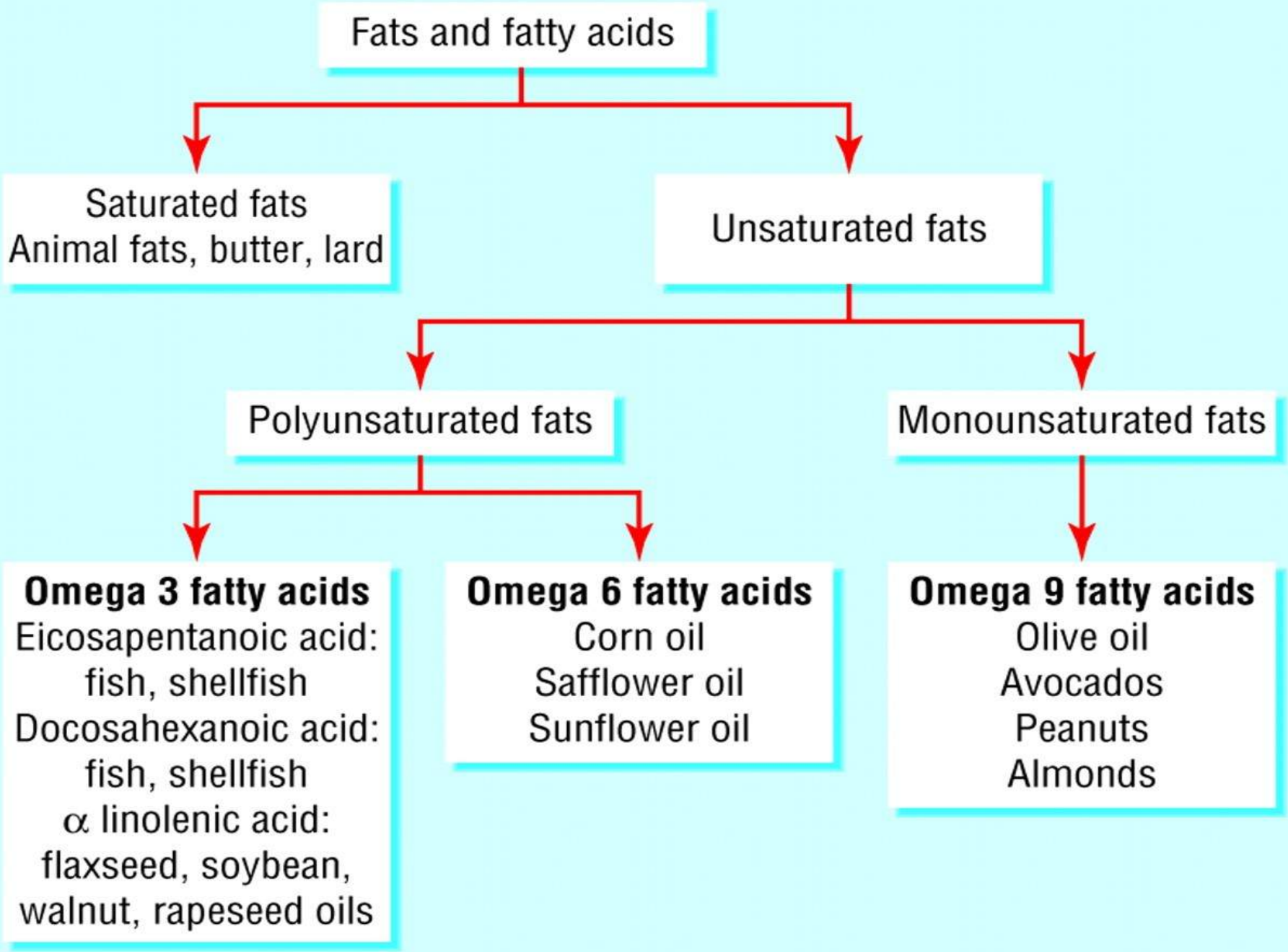
Hydrogenation of Unsaturated Oils

- Hydrogenation converts alkenes to alkanes
- So, hydrogenation of unsaturated oils produces saturated fats
- Hydrogenation is typically carried out by bubbling H_2 gas through the heated oil, in the presence of a metal catalyst (such as nickel or platinum)
- Unsaturated oils are usually only partially hydrogenated, so that the product is not completely saturated, giving a soft semisolid fat such as margarine

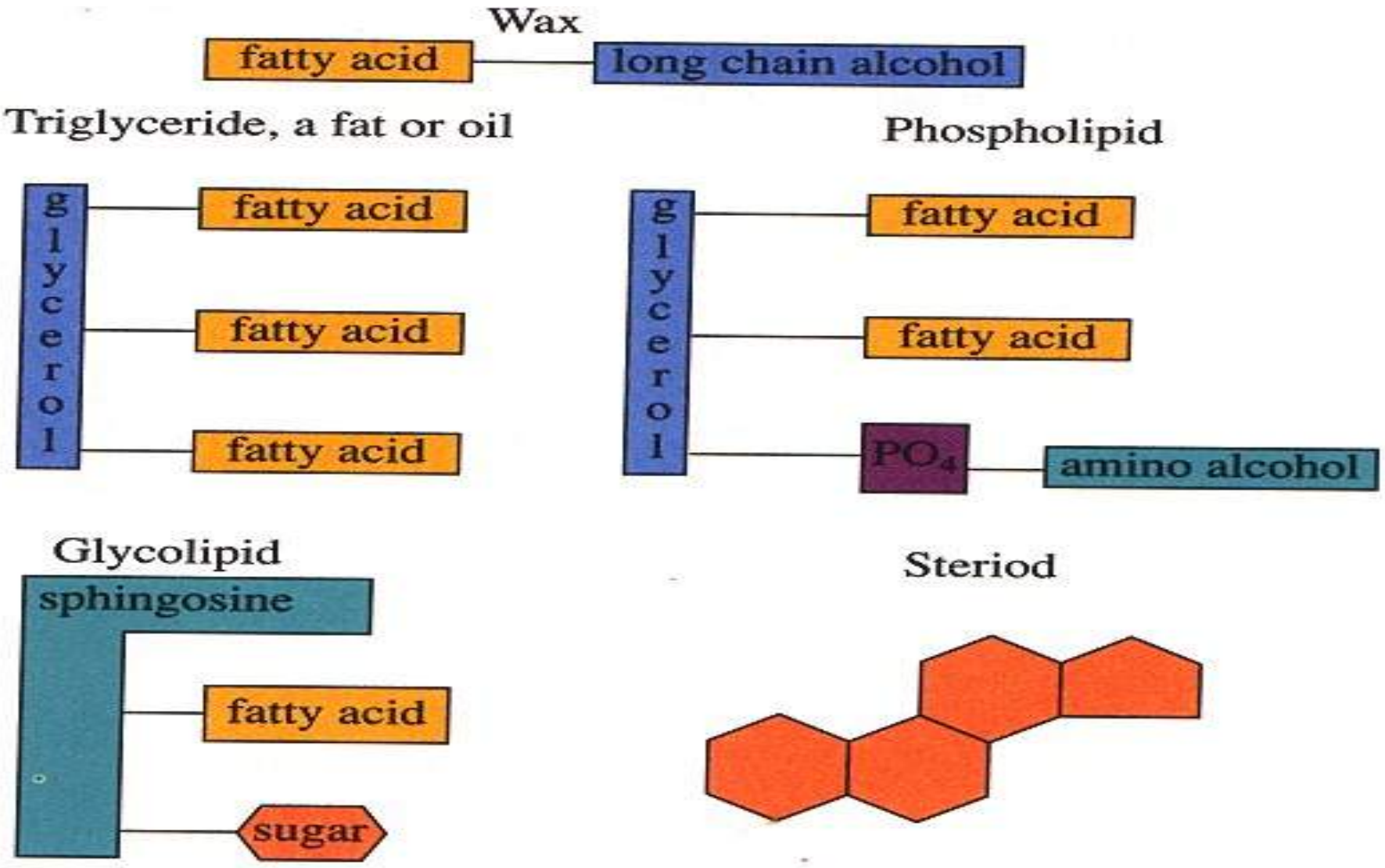
BRANCHED FAs

- Of plant origin, not synthesizedin mammalian organisms
- E.g. *Phytanic acid (in cow milk) – oxidation product of phytol*(part of chlorophyl)

  3,7,11,15 Tetramethylhexadecanoic acid– In sebaceous glands


Fats and fatty acids
Saturated fats
Animal fats, butter, lard
Unsaturated fats
Polyunsaturated fats
Monounsaturated fats
Omega 3 fatty acids
Eicosapentanoic acid:
fish, shellfish
Docosahexanoic acid:
fish, shellfish
α linolenic acid:
flaxseed, soybean,
walnut, rapeseed oils
Omega 6 fatty acids
Corn oil
Safflower oil
Sunflower oil
Omega 9 fatty acids
Olive oil
Avocados
Peanuts
Almonds

Family of Lipids



Nomenclature

- Fatty acids are synthesized from linear acetogenins of acetate pathway

- Fatty acids are carbon chains with a methyl group at one end of the molecule (designated omega, ω) and a carboxyl group at the other end.

- The carbon atom next to the carboxyl group is called the α

carbon, and the subsequent one the β carbon

- The systematic nomenclature for fatty acids may also indicate the location of double bonds with reference to the carboxyl group (Δ).

- $CH_3 - (CH_2)_n CH_2 - CH_2 - COOH$

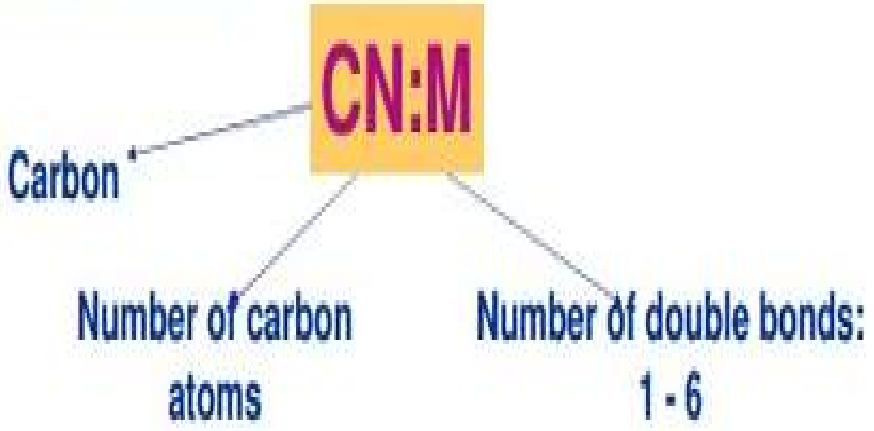
  ω β α

- Fatty acids may be named according to systematic or trivial nomenclature.

- One systematic way to describe fatty acids is related to the methyl (ω) end. This is used to describe the position of double bonds from the end of the fatty acid.

"Telegraphic" nomenclature of FAs (1)

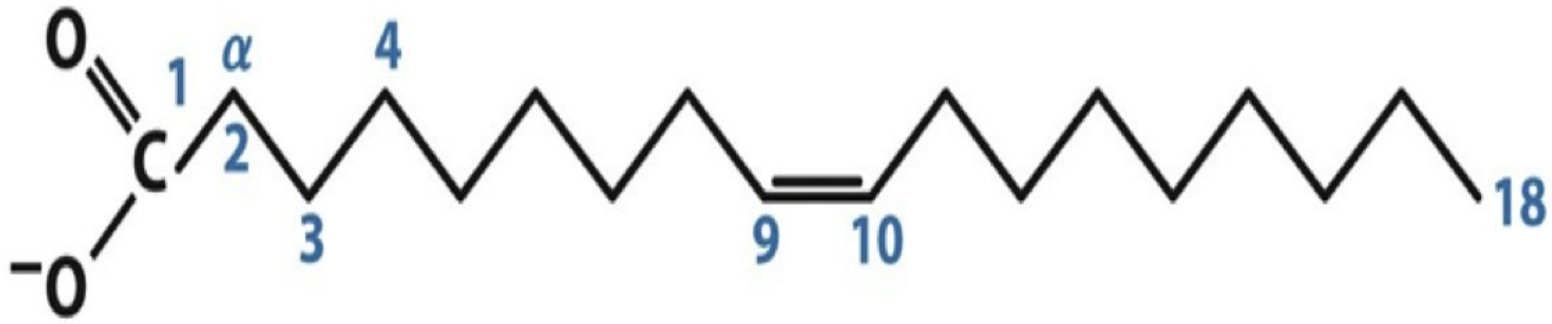
SATURATED FAs:



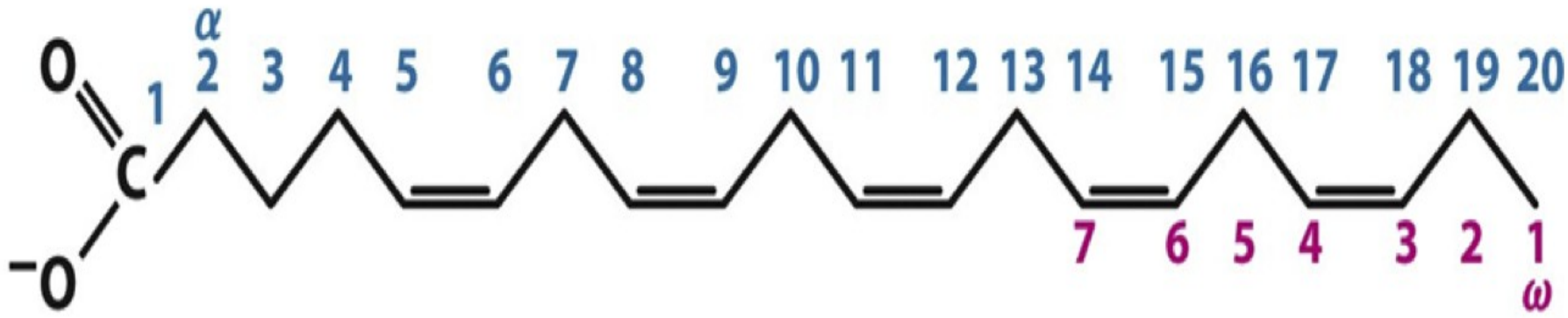
SATURATED FAs WITH A SHORT CHAIN

- If free smell badly
- Fluid
- Present in milk fat (C4-C10)
- Easily digestible

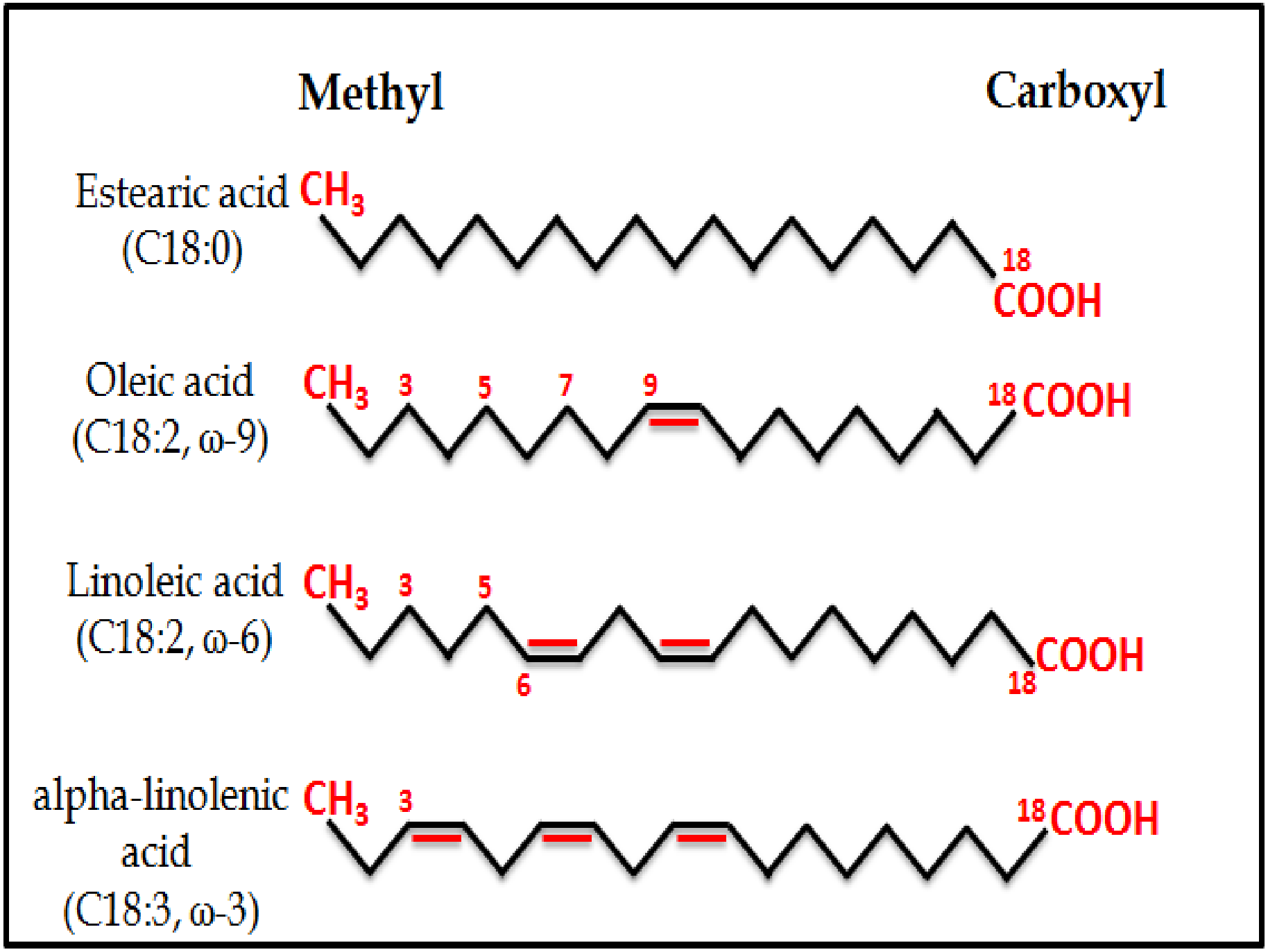
Common name	Systematic name	Formula
C4:0 Butyric acid	Butanoic acid	$CH_3(CH_2)_2COOH$
C6:0 Capronic acid	Hexanoic acid	$CH_3(CH_2)_4COOH$
C8:0 Caprylic acid	Octanoic acid	$CH_3(CH_2)_6COOH$
C10:0 Caprinic acid	Decanoic acid	$CH_3(CH_2)_8COOH$



(a) 18:1(Δ^9) *cis*-9-Octadecenoic acid



(b) 20:5($\Delta^{5,8,11,14,17}$) Eicosapentaenoic acid (EPA), an omega-3 fatty acid


Methyl
Carboxyl
Estearic acid
(C18:0)
CH3
18
COOH
Oleic acid
(C18:2, ω-9)
CH3
3
5
7
9
18
COOH
Linoleic acid
(C18:2, ω-6)
CH3
3
5
6
18
COOH
alpha-linolenic
acid
(C18:3, ω-3)
CH3
3
18
COOH

Uses of fatty acids

- Fatty acids are widely used as inactive ingredients in drug preparations, and the use of lipid formulations as the carriers for active substances is growing rapidly.

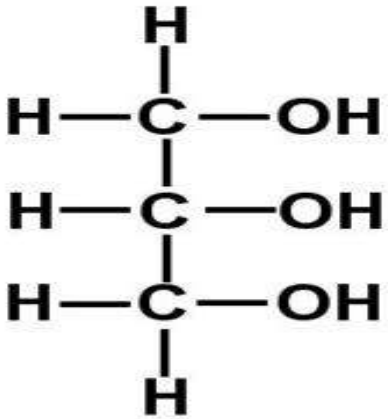
- The largest amount of lipids used in pharmaceuticals is in the production of fat emulsions, mainly for clinical nutrition but also as

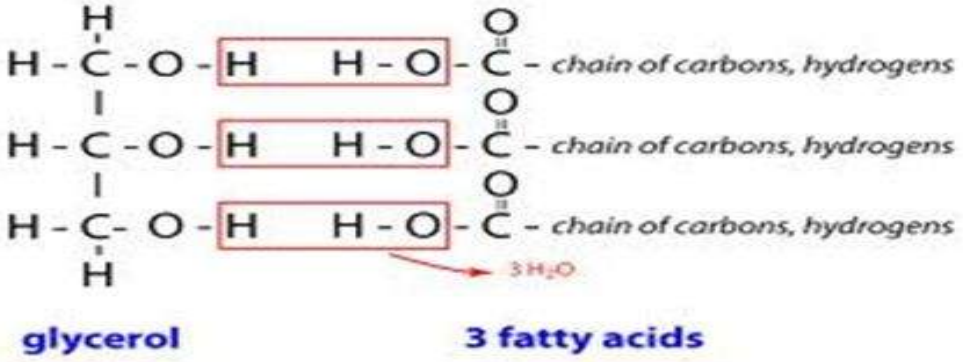
- Another lipid formulation is the liposome, which is a lipid carrier particle for other active ingredients.
- In addition, there has been an increase in the use of lipids

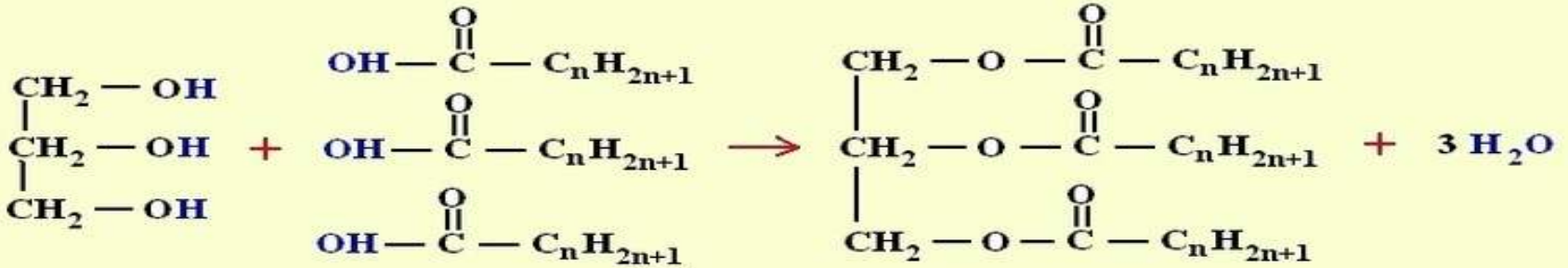
as formulation ingredients.

Triacylglycerols

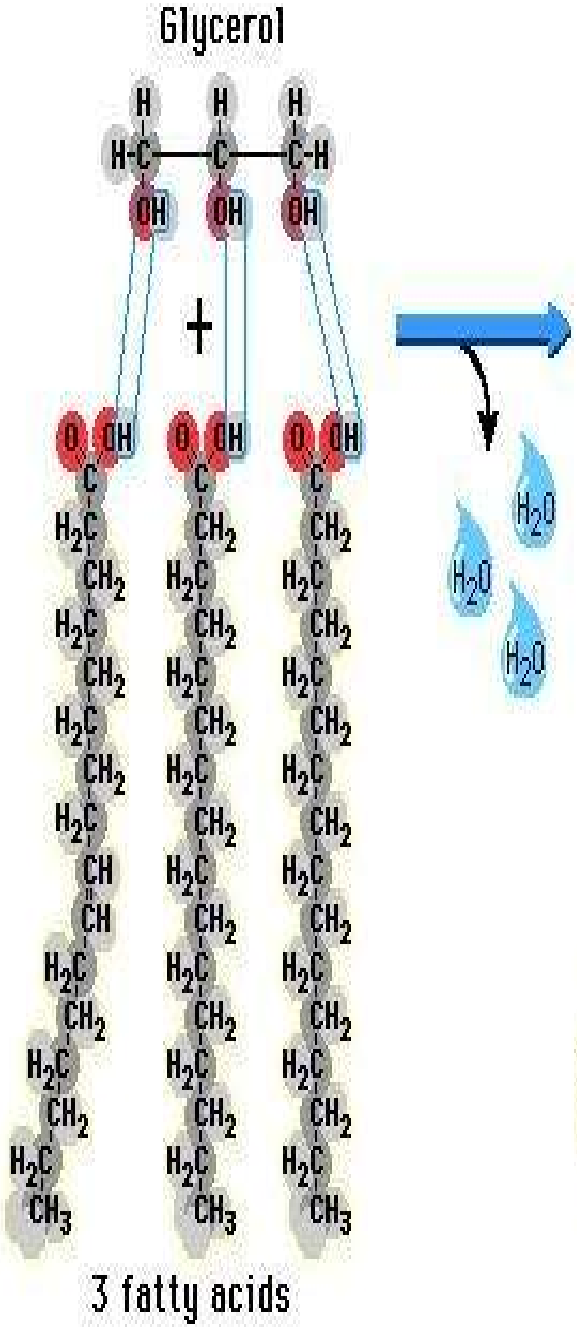
- Glycerol – compound that have 3 OH group
- Triacylglycerol – 3 alcohol group form ester linkages with fatty acid

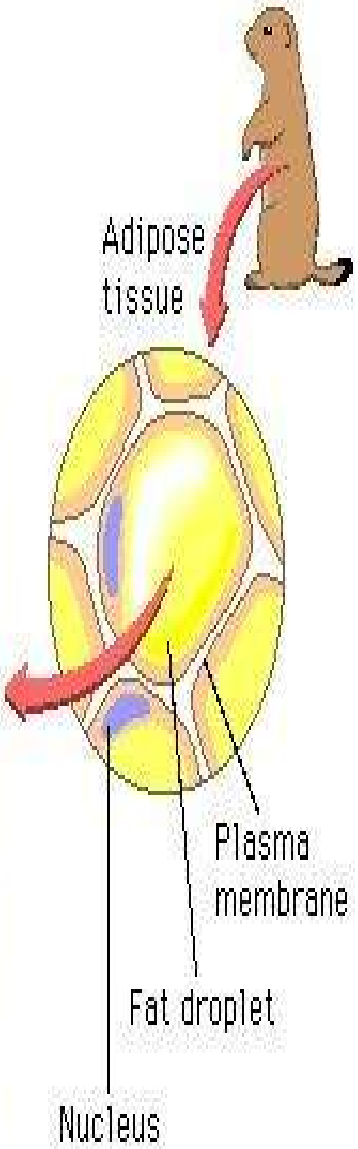







VERY
!HIGH!
Triglycerides
HEALTH RISK


Glycerol
3 fatty acids
Triglyceride
H_2O
H_2O
H_2O

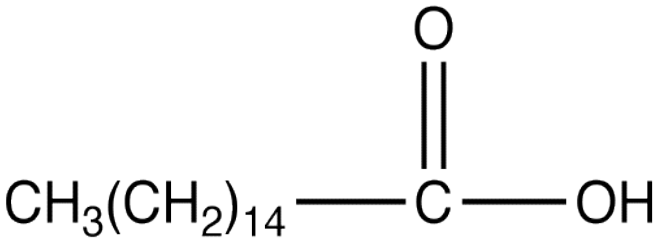

Adipose tissue
Plasma membrane
Fat droplet
Nucleus

Waxes

- Waxes are long-chain saturated and unsaturated fatty acid esters with mono-hydroxy alcohols,
- Have high molecular weight.
- Produced naturally by skin glands as a protection, to keep it lubricated, pliable, and water-proof.

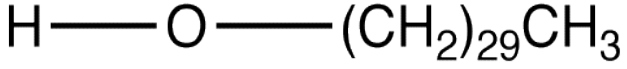






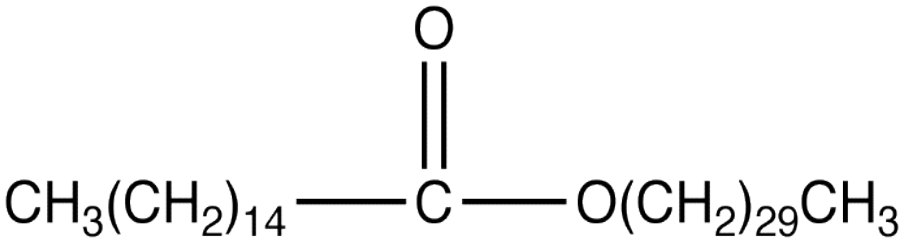
fatty acid

+



long-chain alcohol





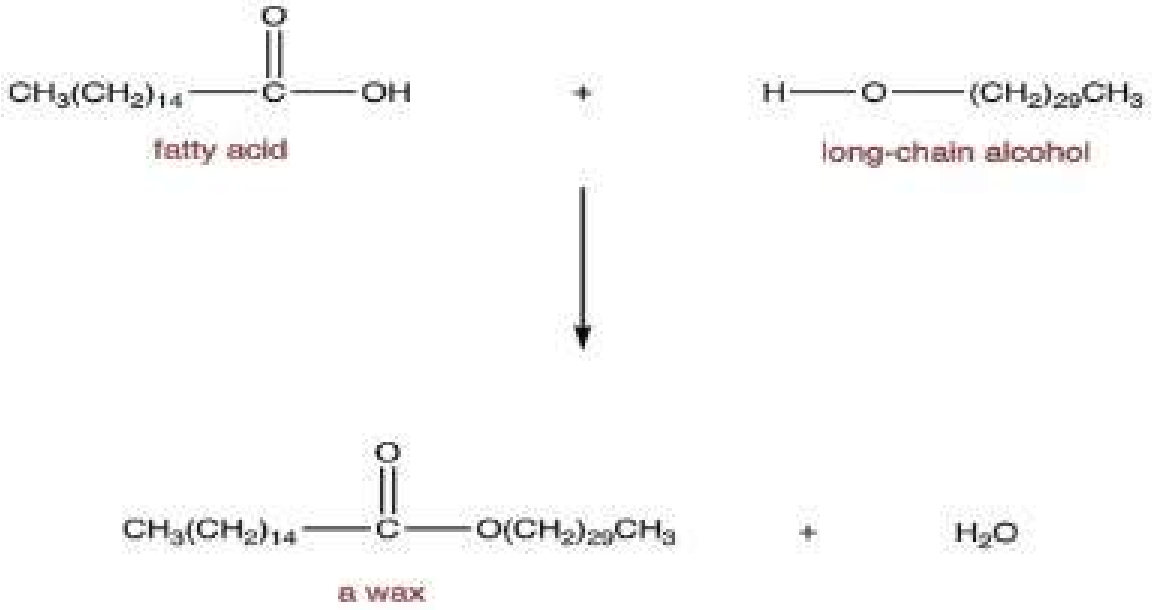
a wax

+

H_2O

Waxes

- A long **fatty acid** chain joined to a long **alcohol** chain
- Waterproof
- Form a protective coating in animals & plants



- common FA's:

  n = 12: lauric acid (n-dodecanoic acid; $C_{12:0}$)

  n = 14: myristic acid (n-tetradecanoic acid; $C_{14:0}$)

  n = 16: palmitic acid (n-hexadecanoic acid; $C_{16:0}$)

  n = 18; stearic acid (n-octadecanoic acid; $C_{18:0}$)

  n = 20; arachidic (eicosanoic acid; $C_{20:0}$)

  n= 22; behenic acid

  n = 24; lignoceric acid

  n = 26; cerotic acid

Compound lipids or Heterolipids

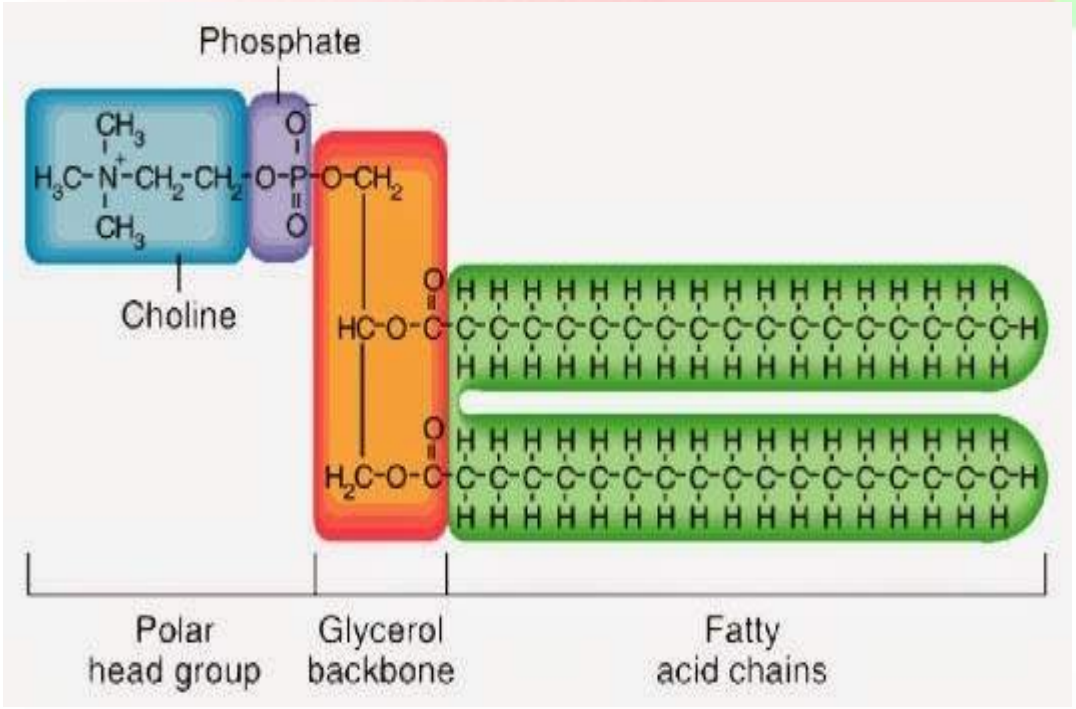
- Heterolipids are fatty acid esters with alcohol and additional groups.
- This category of lipids includes;

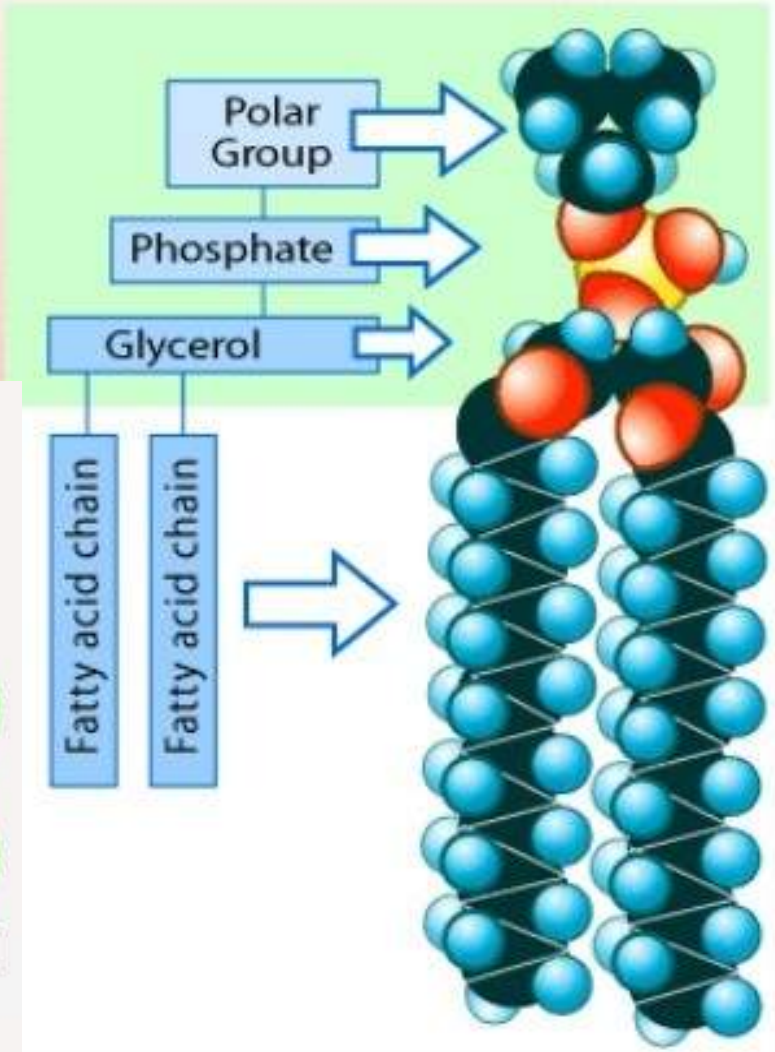
1.Phospholipids

2.Glycophospholipids

Phospholipids

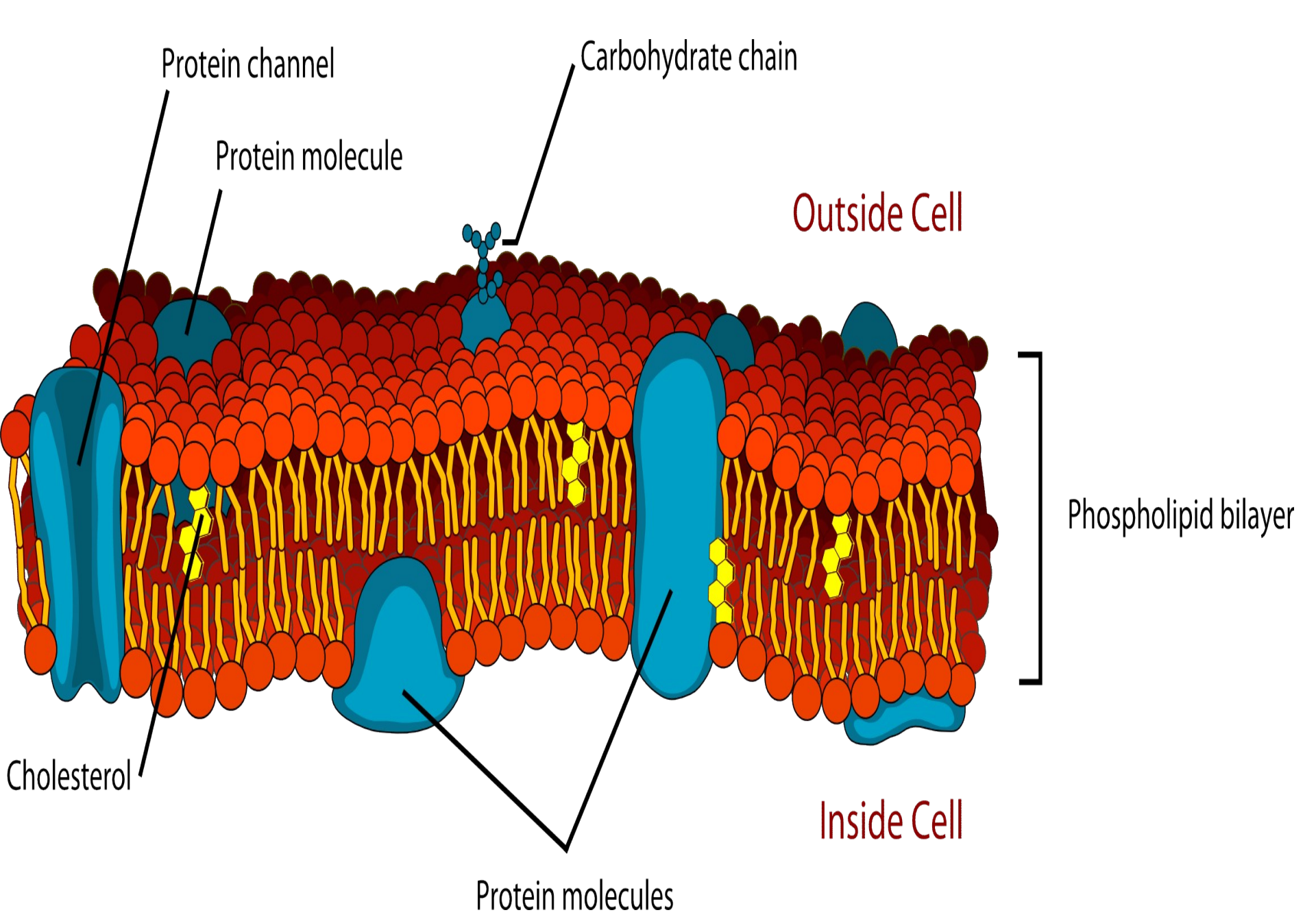
- If one of the fatty acid chains in a triglyceride is replaced with a phosphate group (PO4) a phospholipid is produced





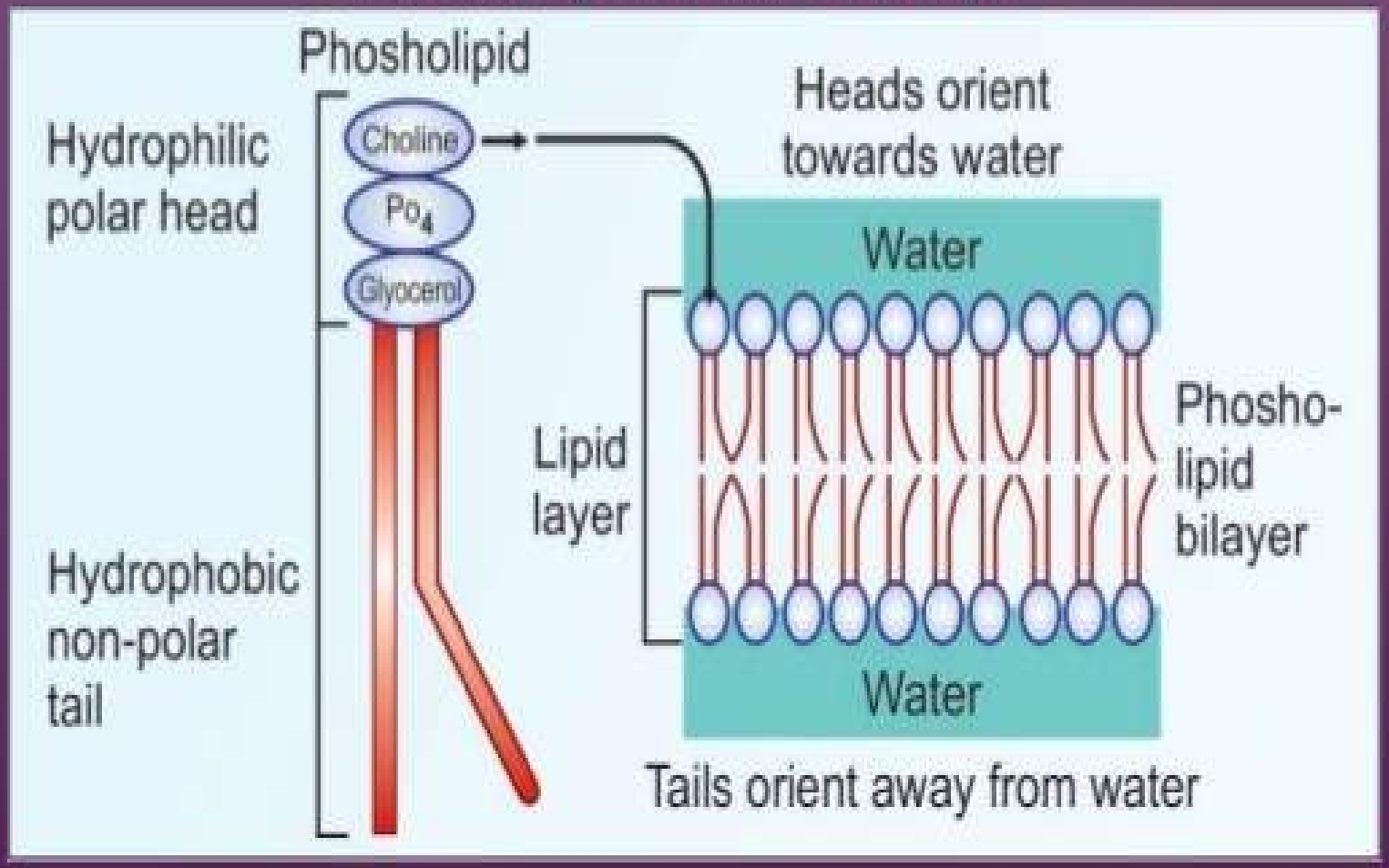
Phospholipids (PL)

- PL contain *fatty acids*, *glycerol*, *nitrogen bases*, *phosphoric acid*, and other substituents.
- They are *most abundant in cell membranes* and serve as structural components.
- They are not stored in large quantities.
- Molecular structure - ***polar***, consisting of one hydrophilic head group and two hydrophobic tails


Protein channel
Carbohydrate chain
Protein molecule
Outside Cell
Phospholipid bilayer
Cholesterol
Inside Cell
Protein molecules

Amphipathic Nature

Phospholipids form the bilayer



Functions of phospholipids

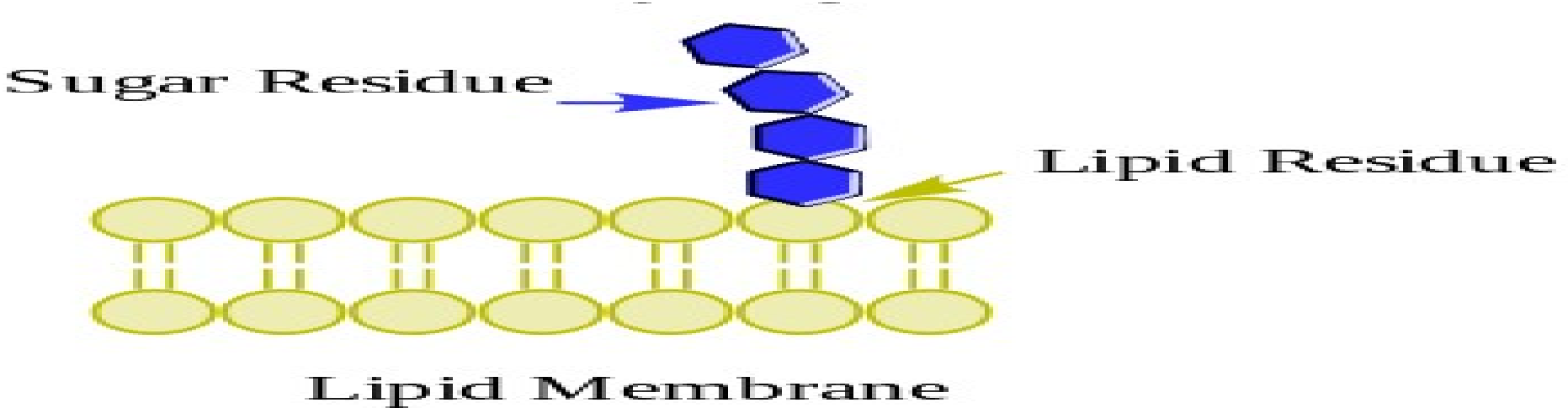
1. In association with proteins phospholipids form the structural components of membranes and regulate membrane permeability
2. Phospholipids in the mitochondria maintain the conformation of electron transport chain components and thus cellular respiration
3. They participate in the absorption of fats from the intestine
4. They are essential for the synthesis of different lipoproteins and thus participate in transport of lipids
5. The prevent accumulation of fats in liver (lipotropic factors)

COMPLEX LIPIDS

Functions of Glycolipids

- Glycolipids are Structural components of plasma membrane





Glycolipids (cerebrosides)

- Glycolipids are fatty acids with carbohydrates and nitrogen but without phosphoric acid.
- Include some compounds like *sulfolipids*, *gangliosides*, and *sulfatids* which are structurally-related.
- These cerebrosides are important constituents of the brain and other tissues. They consist of at least one sugar unit, so they are also called glycosphingosides.
- They are like phospholipids because they have a hydrophobic region, with a polar region and two long hydrocarbon tails.
- Like phospholipids, glycolipids form lipid bilayers that are self-sealing and form the structure of cellular membranes.

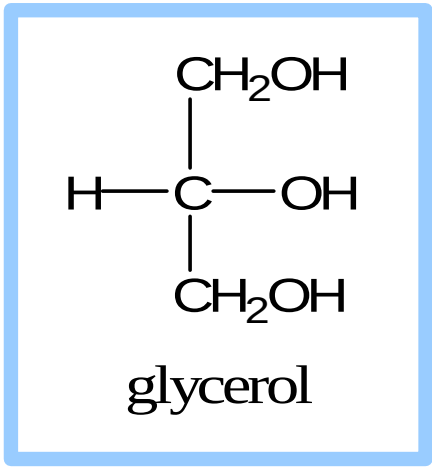
Glycerophospholipids

Glycerophospholipids (phosphoglycerides), are common constituents of cellular membranes.

They have a **glycerol** backbone.

Hydroxyls at **C1** & **C2** are esterified to **fatty acids**.

An **ester** forms when a hydroxyl reacts with a carboxylic acid, with loss of H2O.

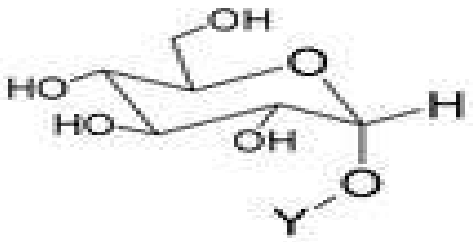


glycerol

Functions of Glycolipid

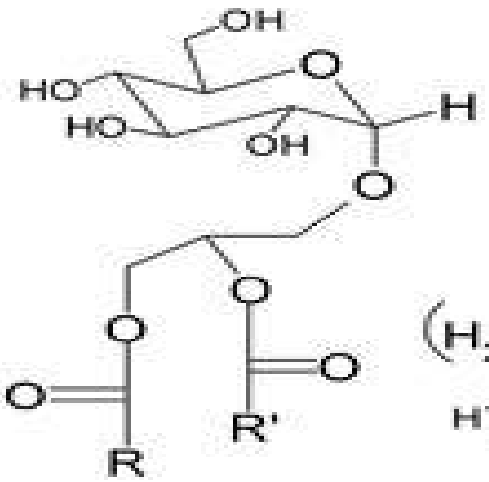
- The glycolipids are an essential part of cell membranes.
- Glycolipids also help determine the blood group of an individual.
- Glycolipids act as receptors at the surface of the red blood cell.
- some viruses, bacteria (eg., cholera) use glycolipids on their cell surface as well. This helps the immune system destroy and clear the pathogen from the body.

Glycolipids

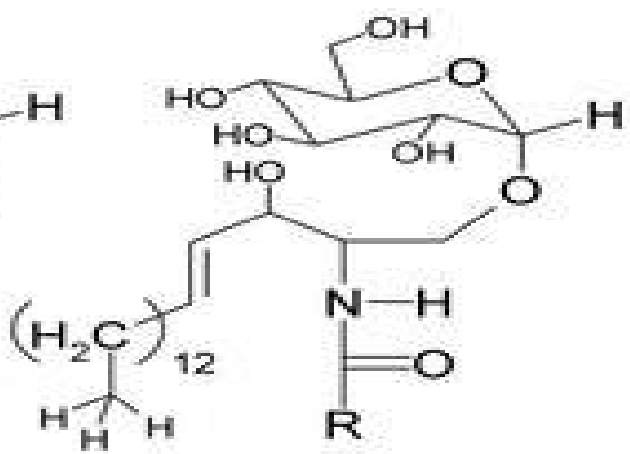


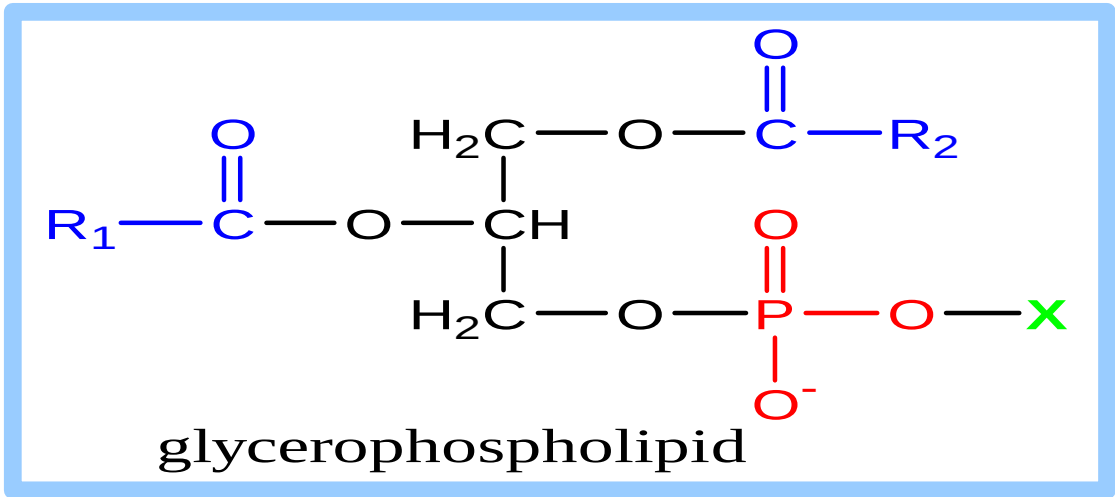
Y = Lipid

Glycero-Glycolipids



Sphingo-Glycolipids

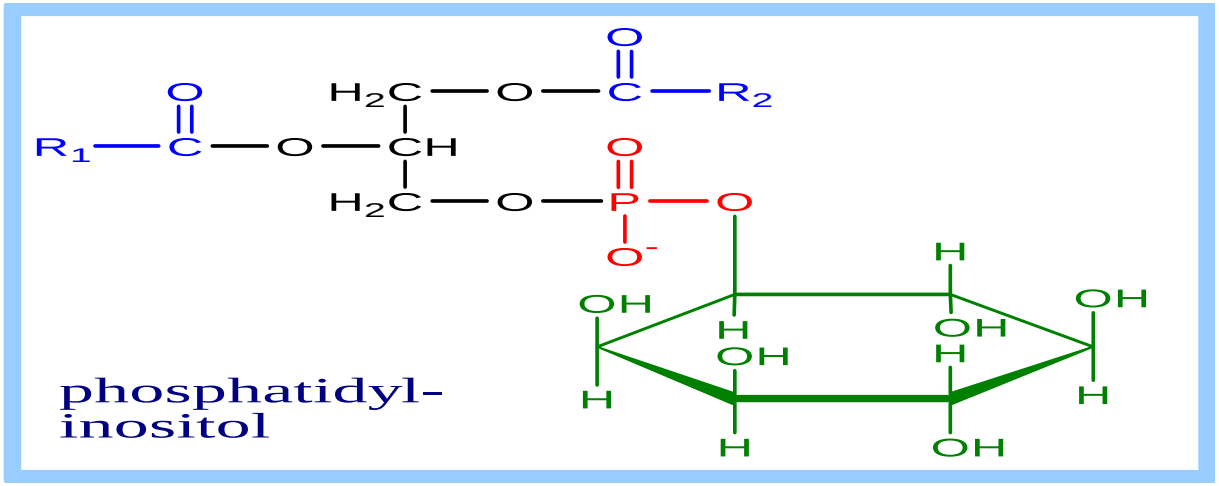




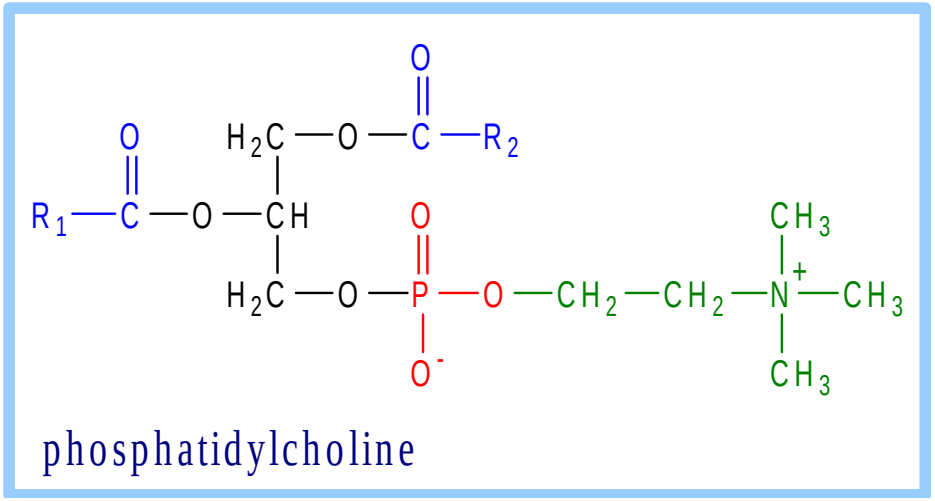
In most **glycerophospholipids** (phosphoglycerides),

Pi is in turn esterified to **OH** of a **polar head group** (**X**): e.g., **serine, choline, ethanolamine, glycerol, or inositol**.

The 2 fatty acids tend to be non-identical. They may differ in length and/or the presence/absence of double bonds.



Phosphatidylinositol, with inositol as polar head group, is one glycerophospholipid.
In addition to being a membrane lipid, phosphatidylinositol ***has roles in cell signaling.***

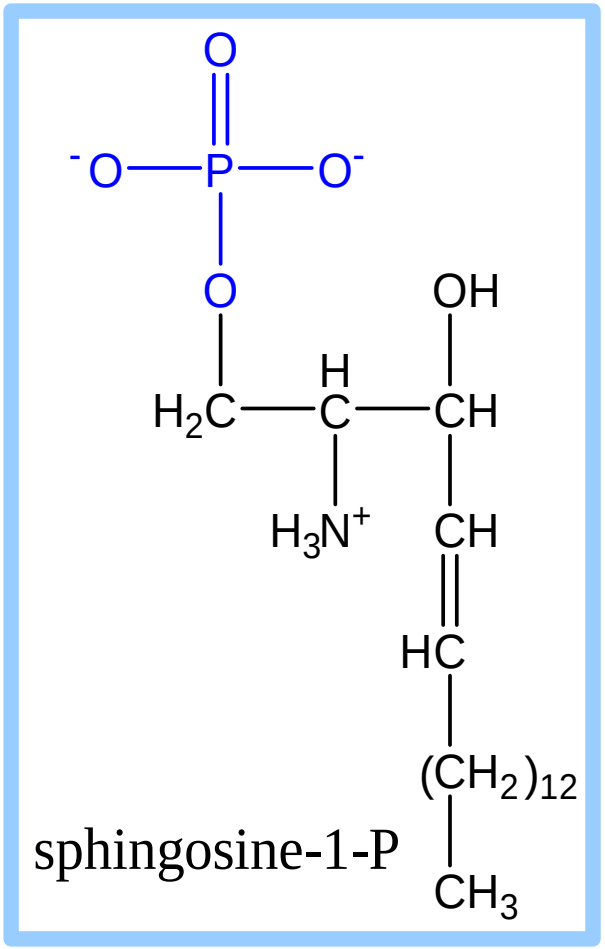


Phosphatidylcholine, with choline as polar head group, is another glycerophospholipid. It is a ***common membrane lipid.***

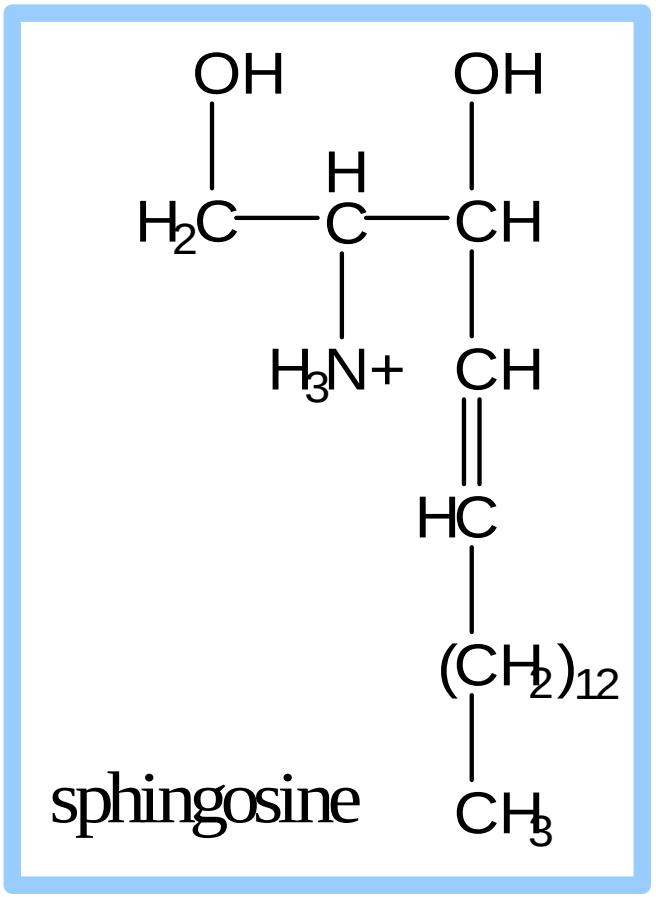
- **Sphingolipids** are derivatives of the lipid **sphingosine**, which has a long hydrocarbon tail, and a polar domain that includes an amino group.

Sphingosine may be reversibly phosphorylated to produce the **signal** molecule **sphingosine-1-phosphate**.

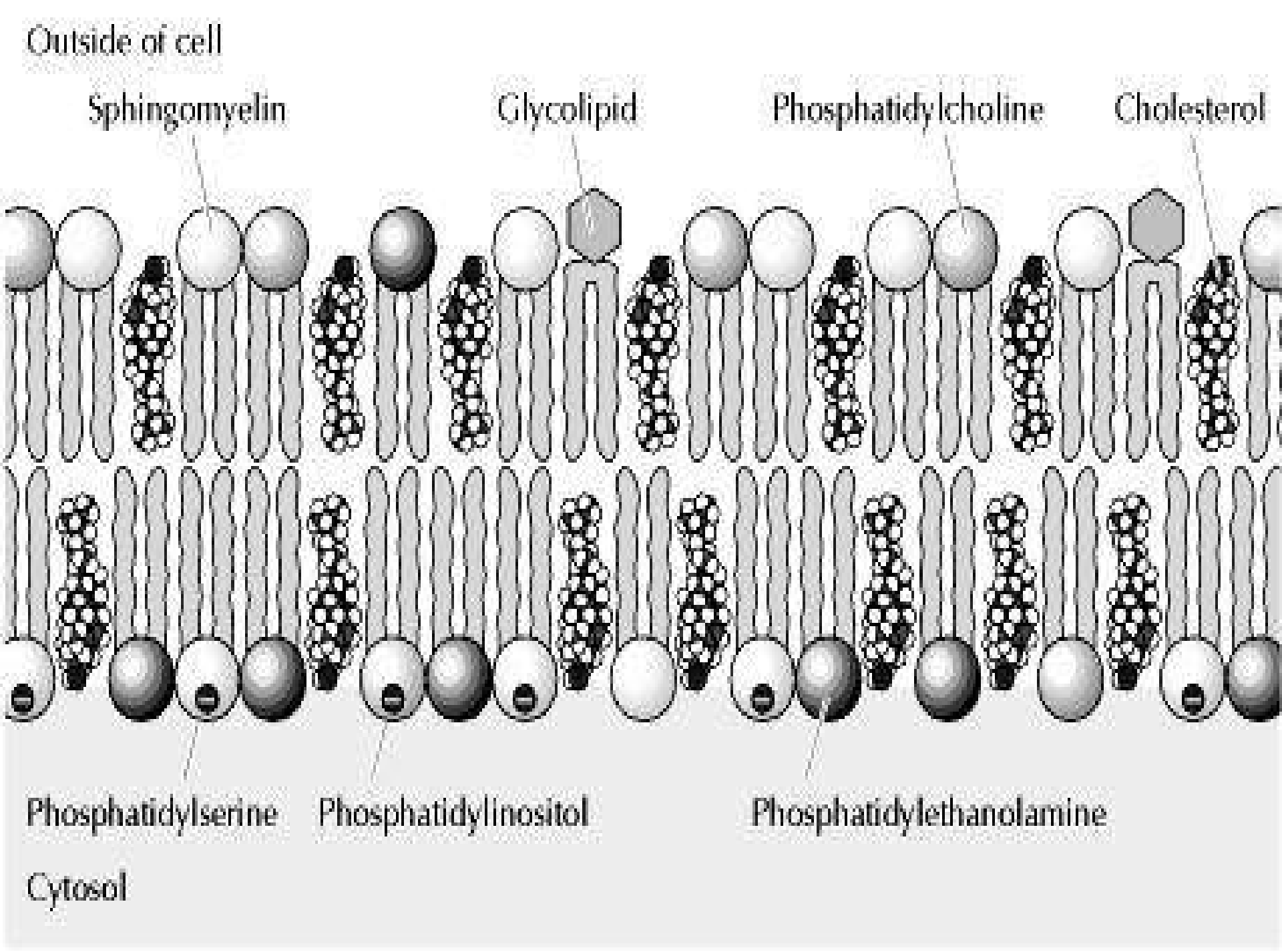
Other derivatives of sphingosine are commonly found as constituents of biological membranes.



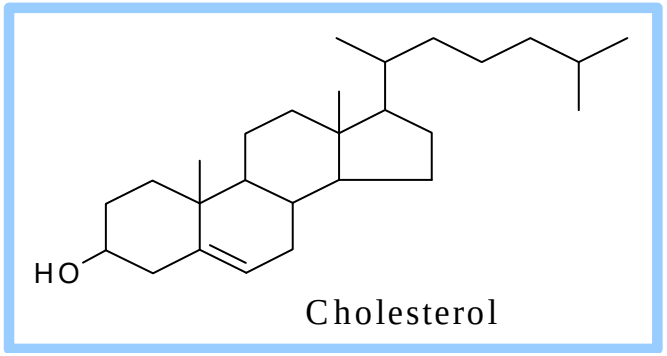
sphingosine-1-P



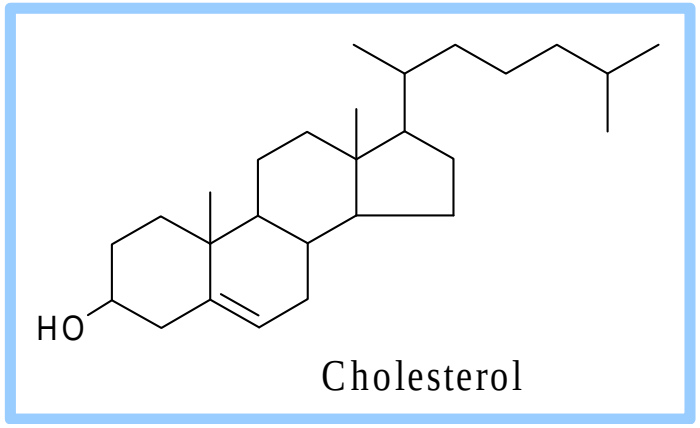
sphingosine


Outside of cell
Sphingomyelin
Glycolipid
Phosphatidylcholine
Cholesterol
Phosphatidylserine
Phosphatidylinositol
Phosphatidylethanolamine
Cytosol

Cholesterol, an important constituent of cell membranes, has a **rigid** ring system and a short branched hydrocarbon tail. Cholesterol is largely **hydrophobic**.
But it has one polar group, a **hydroxyl**, making it **amphipathic**.



Cholesterol





Cholesterol inserts into bilayer membranes with its hydroxyl group oriented toward the aqueous phase & its hydrophobic ring system adjacent to fatty acid chains of phospholipids.

The **OH** group of cholesterol forms hydrogen bonds with polar phospholipid head groups.

Interaction with the relatively **rigid cholesterol** decreases the mobility of hydrocarbon tails of phospholipids.

But the presence of **cholesterol** in a phospholipid membrane interferes with close packing of fatty acid tails in the crystalline state, and thus **inhibits transition** to the crystal state. Phospholipid membranes with a high concentration of cholesterol have a **fluidity intermediate** between the liquid crystal and crystal states.



- Cholesterol
  - Synthesized in the liver
  - Gives strength to cell membranes
  - In myelin sheath, bile salts, vitamin D produced by the skin
  - Excess in body leads to gall stones and plaque in the arteries

Steriods

- Bile salts
  - Synthesized from cholesterol in the liver
  - Sent to the digestive tract to emulsify fats
  - Promotes absorption of cholesterol in the digestive tract
- Fat-soluble vitamins: A, D, E, K
  - A for night vision
  - D promotes absorption of Ca^{+} in intestinal tract and deposition of Ca^{+} in the bones
  - E prevents oxidation of unsaturated fatty acids in cell membranes
  - K functions in blood cloting

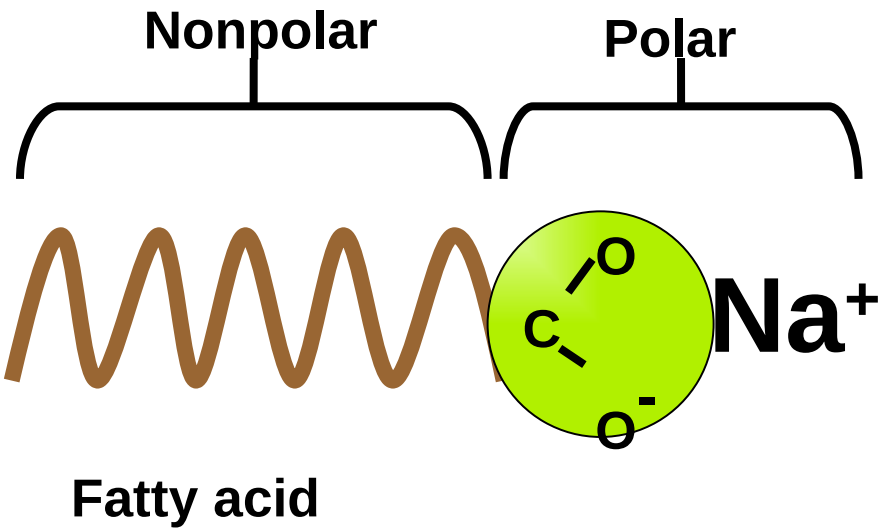
- Hormones:
  - Sex hormones control reproduction, sexual characteristics, and general growth
  - Adrenal corticosteroids
    - Cortisone increases blood glucose levels and reduces inflammation
    - Aldosterone acts in the kidneys to maintain Na^+ and water balance
    - Sex hormones
- Anabolic steroid, derivative of testosterone
  - Prevents breakdown of worn out muscle cells
  - Has dangerous side effects

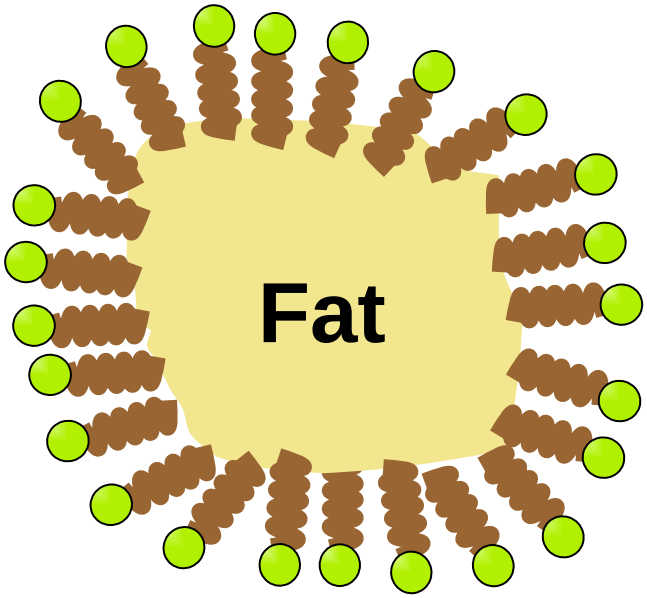
Saponification

- Process of formation of soap is known as saponification
  - Types of soaps:
    - Sodium soap – ordinary hard soap
    - Potassium soap – soft soap (shaving soaps are potassium soaps of coconut and palm oils)
    - Castile soap – sodium soap of olive oil
    - Green soap – mixture of sodium and potassium linseed oil
    - Transparent soap – contains sucrose
    - Floating soap – contains air
    - Calcium and magnesium soaps are very poorly water soluble (hard water contains calcium and magnesium salts –these insolubilize soaps)

Soap:

Long Chain Fatty Acid Salt





Fat with soap is water soluble

Hydrogenated fats

- hydrogenation leads to either saturated fats and or trans fatty acids
- the purpose of hydrogenation is to make the oil/fat more stable to oxygen and temperature variation (increase shelf life)
- example of hydrogenated fats: Crisco, margarine.
- Properties:
  - Hydrogenation (adding H_2)
    - Converts a double bond to a single bond
    - Changes liquid oil to a soft semisolid fat

Eicosanoids

- Eicosanoids are produced from ***Arachidonic acid*** which is a twenty (20) carbon poly unsaturated fatty acid (PUFA).
- Eicosanoids are family of powerful, hormone-like compounds produced in the body from Essential Fatty Acids.
- Eicosanoids include *prostaglandins*, *prostacyclins*, *leukotrienes*, and *thromboxanes*, which are responsible for many of the beneficial effects of the good fats.

Classification and Functions of Eicosanoids

- ***Prostaglandins*** – mediate pain sensitivity, fever production, prevent platelet aggregation, swelling and inflammation. Also use to induce labor.
- ***Thromboxanes*** – produced by platelets, involved in blood clotting, constriction of arteries.
- ***Leukotrienes*** – attract white cells, involved inflammatory diseases (asthma etc) / allergic reaction.

- Arachidonic acid is converted into prostaglandins which serve as local chemical messengers